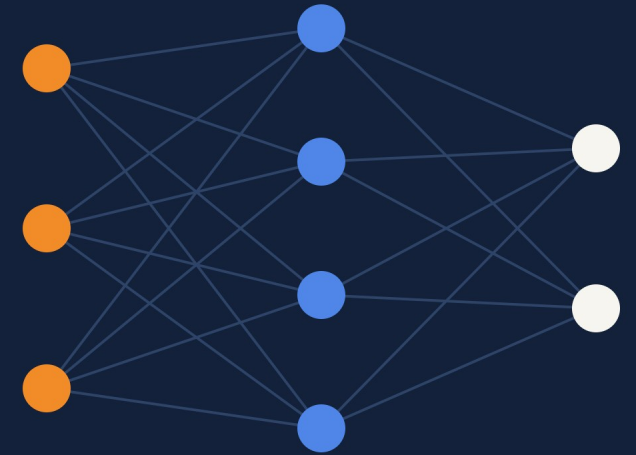


MACHINE LEARNING BOOTCAMP · DAY 1 OF 5

Introduction to Machine Learning



What ML is, how machines learn, and where it's changing the world

Soyinka Enoch · 28/09/2026

Our 5-day journey

Day 1 · Today	The big picture	What ML is, types of learning, Python
Day 2	Data & supervised	Pandas, EDA, regression, classification
Day 3	More models + clustering	Random forests, SVM, K-Means, PCA
Day 4	Deep learning & vision	Neural networks, CNNs, object detection
Day 5	ML in the real world	ESP32, biomedical, robotics, RL, GenAI

Every day: ~1 hr concepts · ~1.5 hrs hands-on · ~30 min applications & Q&A

Today's plan — 3 hours

Time	Block	What we'll do
0:00 – 0:15	Welcome	The 5-day roadmap
0:15 – 0:55	What is ML?	AI vs ML, how machines learn
0:55 – 1:35	Types of learning	Five types, the ML workflow
1:35 – 1:45	Break	Stretch, questions
1:45 – 2:15	Applications	Vision, health, devices, GenAI
2:15 – 2:50	Python & tools	Colab and your first model
2:50 – 3:00	Wrap-up	Responsible ML, recap

AI, ML, deep learning & GenAI

Artificial Intelligence

Machine Learning

Deep Learning

Generative AI

AI — any technique that makes machines act "smart", even fixed rules.

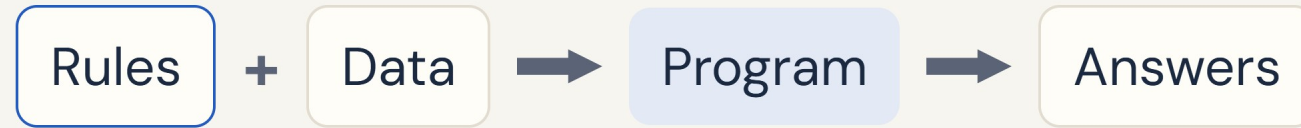
ML — systems that learn patterns from data.

Deep learning — ML with many-layered neural networks.

GenAI — models that create new text, images, audio and code.

Traditional programming vs ML

TRADITIONAL PROGRAMMING



MACHINE LEARNING



In ML we give the computer examples with answers, and it works out the rules itself.

Example: show it thousands of emails labelled spam / not spam.

A short history of ML

1950

Turing asks:
can machines
think?

1957

Rosenblatt
builds the
Perceptron

1959

Samuel coins
"machine
learning"

1997

Deep Blue
beats
Kasparov at
chess

2012

AlexNet starts
the deep
learning boom

2016

AlphaGo
beats Go
champion Lee
Sedol

2022

ChatGPT
brings GenAI
to millions

The ideas are old. What changed recently is data and computing power.

Why is ML exploding now?



Data

Phones, sensors, apps and the internet create huge datasets every day.



Compute

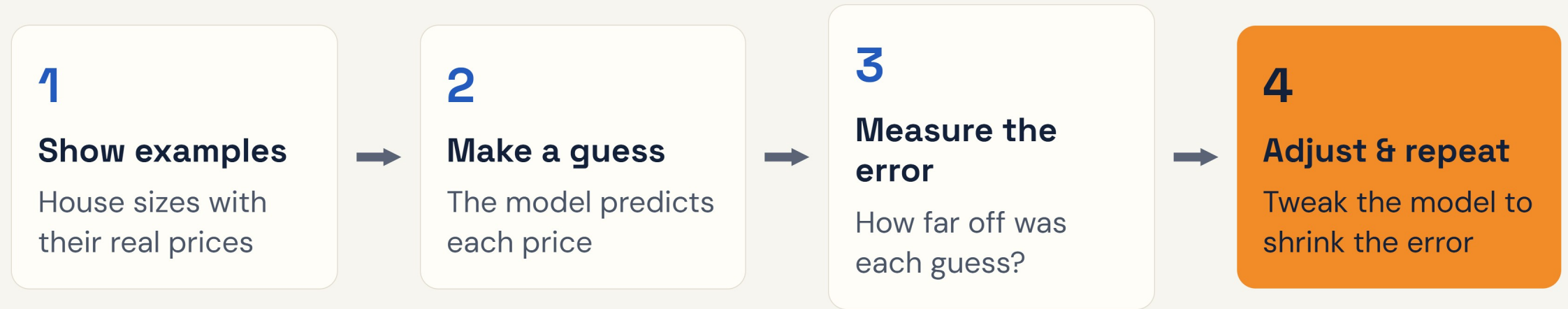
GPUs make training fast and cheap — even free on Google Colab.



Tools

Free libraries: scikit-learn, TensorFlow, PyTorch.

How does a machine learn?



Repeat thousands of times → the error shrinks. That's "training".

Like exam prep: **past questions** → **marking scheme** → **learn from mistakes** → **try again**.

Key words you'll hear all week

Features (inputs)			Label
m ²	Rooms	Area	Nm
120	3	Lekki	85
80	2	Yaba	45
200	4	Ikoyi	250
95	2	Surulere	?

Made-up numbers. The last row is a new house — the model predicts its price.

Feature — an input column (size, rooms, area)

Label — what we predict (price)

Sample — one row, one example

Dataset — all the rows together

Model — the learned pattern

Train / test — learn on some rows, check on unseen rows

PART 2

Five ways machines learn

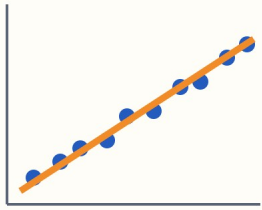
Supervised · Unsupervised · Semi-supervised · Self-supervised
· Reinforcement

Supervised: learning with answers

We give the model inputs **and** the correct answers (labels).

Regression

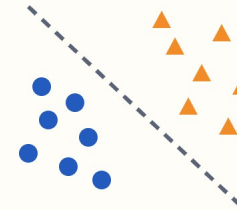
Predict a number



- House price in naira
- Tomorrow's temperature
- Crop yield per hectare
- Battery life remaining

Classification

Predict a category



- Spam or not spam
- Cancerous or not
- Handwritten digit 0–9
- Fruit ripe or unripe

Days 2–3: linear & logistic regression, KNN, decision trees, random forests, SVM

Unsupervised: find hidden patterns

No labels. The model finds structure in the data on its own.



Clustering

Group similar things

- Customer groups
- Similar news articles
- Songs by style



Dimensionality reduction

Many columns → a few

- 64-pixel images → 2D
- Compress sensor data



Anomaly detection

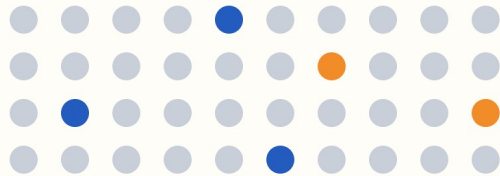
Spot the odd one out

- Fraud alerts
- A motor vibrating oddly
- An abnormal heartbeat

Semi- & self-supervised learning

Semi-supervised

Few labels + lots of unlabelled data



Why? Labelling is costly — a doctor can't label 10,000 X-rays.

How: learn from the few → label the rest
→ retrain.

Self-supervised

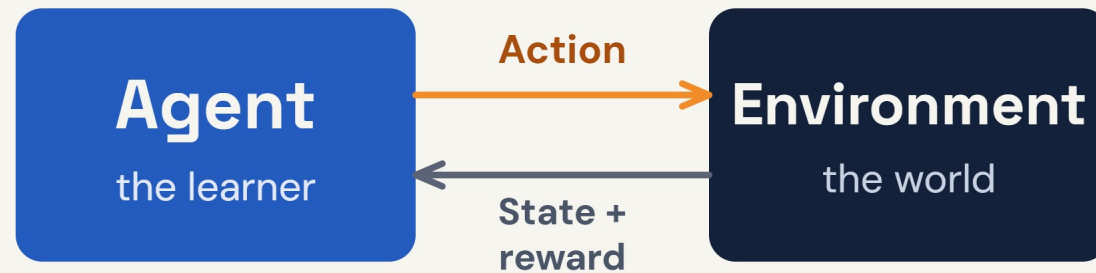
The data creates its own labels

"The students _____ into the
lecture hall."

Model's guess: **walked**

Hide part of the data and predict it. This is how ChatGPT first learns.

Reinforcement: learning by reward



Agent — the decision-maker

Environment — the world it acts in

Action — what it chooses to do

Reward — points for good moves, penalties for bad

Policy — its strategy: "here, do this"

Like training a dog: a treat for good behaviour, nothing for bad.

Examples: AlphaGo · robots learning to walk · game AI · warehouse robots

The five types at a glance

Type	Data it needs	Example
Supervised	Labelled examples	Spam filter
Unsupervised	No labels	Customer groups
Semi-supervised	Few labels + many unlabelled	Medical images
Self-supervised	Data that labels itself	ChatGPT pre-training
Reinforcement	Rewards from trial and error	Game AI, robots

The ML workflow

1 Define the problem

What to predict?

2 Collect data

Sensors, surveys, web

3 Explore (EDA)

Plot and understand it

4 Clean the data

Fix gaps and errors

5 Features

Build useful inputs

6 Train the model

Learn from data

7 Evaluate

Test on unseen data

8 Deploy

App, website or ESP32

9 Monitor & improve

Watch it, retrain

Steps 2–5: **Day 2** · Train & evaluate: **Days 2–4** · Deploy: **Day 5**

PART 3

Where is ML used?

Almost everywhere — including the fields you're studying.

ML in engineering fields

Computer vision

Face unlock, self-driving cars, number plates, crop disease from a leaf photo.

Bioinstrumentation

Smartwatches spot irregular heartbeats in ECG. EMG and EEG control prosthetic hands.

Embedded ML

Tiny models on chips like the ESP32: voice commands, gestures, fall detection.

Robotics

Robots learn to navigate, grasp and walk — in simulation first, then for real.

ML you already use every day

Fintech

Fraud alerts, credit scoring for loan apps

Agriculture

Crop disease, yield forecasts, irrigation

Recommendations

Your next video on YouTube, TikTok, Netflix

Language

Translate, voice assistants, autocorrect

Generative AI

ChatGPT, Claude, Gemini, image generators

Transport

Google Maps traffic, Uber and Bolt prices

Your turn: name one ML use.

Spotlight: ML on tiny devices

Why run ML on the device?

Speed — no network round-trip (200 ms or more)

Offline — works where signal is poor

Low power — runs on small batteries

Privacy — data stays on the device

Popular boards

ESP32 / ESP32-CAM

Cheap Wi-Fi chip: sensors, sound, simple vision

Arduino Nano 33 BLE Sense

Built-in sensors: gestures, wake words

Raspberry Pi 4 / 5

Small Linux computer: cameras, object detection



PART 4

Python & the ML toolbox

Just enough Python to follow the notebooks this week.

Why Python for ML?

Readable

Reads almost like English.

Huge ecosystem

A free library for nearly every ML task.

Big community

Most ML tutorials and research use it.

PYTHON

```
scores = [65, 72, 88, 91]
average = sum(scores) / len(scores)
print("Average:", average)
# Average: 79.0
```

Python essentials in 5 minutes

1 · Variables & types

```
name = "Ada" # text (str)
age = 20 # whole number (int)
gpa = 4.5 # decimal (float)
is_student = True # bool
```

2 · Lists & dictionaries

```
temps = [31, 29, 33]
temps.append(30)
print(temps[0]) # 31
student = {"name": "Ada", "level": 200}
```

3 · Loops & conditions

```
for t in temps:
    if t > 30:
        print(t, "hot")
    else:
        print(t, "ok")
```

4 · Functions

```
def to_fahrenheit(c):
    return c * 9/5 + 32

print(to_fahrenheit(30)) # 86.0
```

The ML toolbox for this week

Library	What it does	Day
NumPy	Fast maths on arrays of numbers	2
Pandas	Tables in code (like Excel)	2
Matplotlib & Seaborn	Charts and plots	2
scikit-learn	Regression, classification, clustering	2–3
TensorFlow, PyTorch	Neural networks, deep learning	4
OpenCV, YOLO	Images, video, object detection	4
Gymnasium	Practice worlds for RL	5
Hugging Face	Ready-made language & GenAI models	5
TF Lite, Edge Impulse	Models on ESP32 and Raspberry Pi	5

Our lab: Google Colab

Why Colab?

- Free, in your browser
- Nothing to install
- Free GPU
- Code + notes + charts
- Saves to Google Drive

- 1 Go to **colab.research.google.com**
- 2 Sign in with a Google account
- 3 Open the notebook link I share
- 4 Click a cell, press **Shift + Enter**
- 5 Day 4: **Runtime** → **Change runtime type** → **GPU**

Your first ML model in 10 lines

Iris dataset: 150 flowers, 4 measurements each, 3 species. Can a model tell the species?

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier

X, y = load_iris(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = DecisionTreeClassifier()
model.fit(X_train, y_train)
print("Accuracy:", model.score(X_test, y_test))
```

1 · Load data



2 · Split 80/20



3 · Train (fit)



4 · Test (score)

Using ML responsibly

Bias

Models copy their data. Face systems trained mostly on lighter skin make more errors on darker skin.

Privacy

Health, location and chat data are personal. Collect only what you need.

Garbage in, garbage out

Wrong labels lead to wrong predictions.

Over-trusting AI

Models make confident mistakes; GenAI can invent facts. Always verify.

Recap: what we covered today

- 1 ML learns patterns from data, not hand-written rules
- 2 Learning loop: guess → error → adjust → repeat
- 3 Five types: supervised, unsupervised, semi-, self-supervised, reinforcement
- 4 Workflow: data → EDA → features → train → evaluate → deploy
- 5 ML is everywhere: vision, health, devices, robotics, GenAI

BEFORE DAY 2

- Run the Day 1 notebook in Colab
- Find one ML use in your field
- Revise lists, loops, functions

Tomorrow: NumPy, Pandas, EDA, regression, classification

Questions?

Tomorrow: NumPy, Pandas, EDA and your first real models.

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